

CONTACT

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TECHNICAL SKILLS

Languages
Python, TypeScript, JavaScript, Java, C++, SQL

Frontend
React, Next.js, Tailwind CSS, GSAP, Three.js / React Three Fiber, Zustand

Backend
FastAPI, Node.js, Express, REST APIs

Databases
PostgreSQL / PostGIS, Supabase, MongoDB, Qdrant, Oracle SQL

AI / ML
LangGraph, LangChain, Gemini API, RAG, XGBoost, ONNX, computer vision, sensor forecasting, model calibration

Data
Power BI, DAX, Pandas, NumPy, GIS

Tools & Delivery
Git, GitHub Actions, Docker Compose, Vercel, Netlify, Render, pytest, Playwright

CERTIFICATIONS

- Oracle AI Database SQL Certified Associate
  Oracle University, Aug 2026
- Geographic Information Systems, Elite (68%)
  NPTEL / IIT Roorkee, Jan-Apr 2026
- Introduction to Deep Learning
  Infosys Springboard, Jul 2026
- Java Foundation; Programming using Java; Basics of Python
  Infosys Springboard, 2025
- Software Engineering and Agile Development
  Infosys Springboard, Dec 2024
- AI for Beginners
  HP LIFE, Jul 2025
- Power BI Workshop
  OfficeMaster, Aug 2025

LOKTRISHAL K

AI & Full Stack Developer | B.Tech AI & Data Science

CAREER OBJECTIVE

B.Tech AI & Data Science student (CGPA 8.0) who builds agent workflows, ML pipelines and the web apps around them. Recent work: farm agents behind a safety gate, a seven-language healthcare navigator and an exoplanet classifier that runs in the browser. Looking for an AI or full-stack internship.

EDUCATION

B.Tech | Artificial Intelligence & Data Science — Expected Jan 2028
R.M.K. College of Engineering and Technology, Tamil Nadu | CGPA: 8.0

INTERNSHIPS

Data Science Intern, Virtual Tech Services, Chennai — Jun 2025
- Four-week internship; project: **Diabetes Prediction using Machine Learning**.

Vocational Trainee, Chennai Port Authority — Jun 2026
- Two-week non-stipendiary in-plant training, Electronic Data Processing section, M&E Engineering Dept.

AWARDS & HACKATHONS

- **Innovation Award**, Cyber AI Hackathon 2026 (University of Derby, UK; supported by AMSE), Oct 2026
- **Winner**, PrototypeX, IES Summit 2026 (Jeppiaar Engineering College), Sep 2026
- Participant: ISRO BAH 2026, HackWithHyderabad 3.0, DECODE SIH 2026, DEV HOURS Codeathon

PROJECTS

Solarda: Agentic Smart Agriculture — Agentic AI Hackathon 2026
React, TypeScript, Express, FastAPI, LangGraph, ONNX Runtime, AquaCrop-OSPy, Gemini, Docker
- LangGraph farm workflow where agents only propose actions. A deterministic TypeScript safety engine decides what runs, risky actions wait for a human, the gate fails closed if the API is down, and every action is logged.
- Plant-disease classifier (EfficientNetV2-B0, ONNX) that abstains when unsure: 98.2% on PlantVillage, 61.0% on PlantDoc field photos, and 87% correct on the images it chooses to answer.
- Crop and soil-water twin used about 19-32% less water than a rule-based controller at similar yield (replayed data and simulation, not live sensors).

KinSight: Inherited-Risk Screening Navigator — Future Health Digital Innovation Challenge
Next.js, TypeScript, FastAPI, LangGraph, PostgreSQL, Redis, Docker
- Explains screening results in plain language in seven languages (English, Arabic, Hindi, Tamil, Telugu, Kannada, Malayalam), with right-to-left Arabic layout.
- A deterministic genetics engine produces every number. The couple view opens only after both partners agree (higher-risk cases also need a counsellor). Tamper-evident audit log; all data is synthetic.

ExoDetect: Exoplanet Candidate Detection — ISRO Bhartiya Antariksh Hackathon 2026
Python, XGBoost, ONNX Runtime Web, FastAPI, lightkurve, NASA MAST
- Pipeline from raw Kepler light curves (Savitzky-Golay detrending, BLS period search, 13 transit features) to a star-level 70/15/15 split, so one star never lands in two splits.
- XGBoost with isotonic calibration, a threshold picked on validation only, and TreeSHAP explanations. Exported to ONNX so it runs in the browser. On a 400-star sample, AUC-PR fell to 0.913 once depth and SNR features were removed.

NWIS: Nearby Wells Intelligence System — Hackathon prototype
React, FastAPI, PostGIS, Qdrant, LangGraph, Docker
- Ranks offset wells by similarity instead of distance alone and ties each risk estimate to a source report and page. A four-agent LangGraph flow cites sources or says evidence is insufficient.
- Role-based access, hash-chained audit log; 251 tests pass on a synthetic 12-well dataset, not real field data.

Legal & Regulatory Assurance Platform — In development
FastAPI, React, PostgreSQL, Qdrant, PyMuPDF
- Secure document intake (hashing, quarantine), TXT/DOCX/PDF extraction with page-level locators, and a resolver that returns exact stored quotes. 126 scope and security tests pass; contract analysis still planned.

OTHER PROJECTS

- **ResumeForge AI** *(Next.js, LangGraph, Gemini 2.0, Supabase)*: six-agent resume and career pipeline with OAuth and OTP sign-in.
- **SentinelPay** *(React, Node.js, MongoDB, Gemini API)*: UPI fraud risk scoring, deployed on Vercel with GitHub CI/CD.
- **PeopleRamp / OnboardIQ** *(React, Node.js, Gemini)*: AI onboarding tools with voice, text and camera input.
- **EduNova** *(Next.js 15, React Three Fiber)*: science-learning site with video-quest worlds.
- **Sri Ambal Healthcare** *(HTML/JS, Supabase, Gemini)*: hospital site with booking to Supabase and a chatbot.
- **Amazon Sales Dashboard** *(Power BI, DAX, SQL)*: cleaned raw sales data; built KPI, trend and category views.